

Econ 527 HW4

Wealth Distribution in Incomplete Market Models

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1 The Huggett Model

1.1 Setup

We solve the infinite-horizon Huggett (1993) model. Each household maximises discounted expected utility

$$\max_{\{a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \quad (1)$$

subject to the period budget constraint

$$c_t + a_{t+1} = (1 + r)a_t + \exp(z_t), \quad c_t \geq 0, \quad a_{t+1} \geq a_{\min}, \quad (2)$$

where z_t is idiosyncratic log-income following the AR(1) process

$$\log z_t = \rho \log z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2). \quad (3)$$

The utility function is CRRA with risk-aversion parameter γ :

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}. \quad (4)$$

Household optimality is characterised by the Euler equation

$$u'(c_t) = \beta(1 + r) \mathbb{E}_t[u'(c_{t+1})], \quad (5)$$

which holds with equality whenever the borrowing constraint $a_{t+1} \geq a_{\min}$ does not bind.

Bonds are in zero net supply, so the goods market clears when

$$\sum_i \sum_j a_i \phi(a_i, z_j) = 0, \quad (6)$$

where $\phi(a, z)$ is the stationary joint distribution of assets and income. The equilibrium interest rate r^* is defined as the unique r satisfying (6).

1.2 Calibration

The model is calibrated at annual frequency. Parameters are set as follows:

Parameter	Symbol	Value	Source / Target
Risk aversion	γ	2	Standard
Discount factor	β	0.96	Given
AR(1) persistence	ρ	0.9	Given
AR(1) std. dev.	σ	0.2	Given
Income grid points	n_z	7	Rouwenhorst
Asset grid points	n_a	150	Given
Grid curvature	κ	2	Given
Borrowing limit	$a_{\min}$	$-\bar{z}$	Average income
Upper bound	$a_{\max}$	$50\bar{z}$	$50\times$ average income

Table 1: Huggett model calibration. $\bar{z} = \mathbb{E}[\exp(z)] \approx 1.30$.

The borrowing limit is set to average income, $a_{\min} = -\bar{z} \approx -1.30$, reflecting that households cannot borrow more than one year’s mean income. The AR(1) process is discretised using Rouwenhorst’s method with $n_z = 7$ grid points. The asset grid uses a polynomial spacing with curvature $\kappa = 2$, which concentrates points near the borrowing constraint where the policy function is most curved.

1.3 Algorithm

Value function iteration. We solve the household’s problem using value function iteration (VFI) with Howard policy improvement. Starting from an initial guess $V^0(a_i, z_j) = u(Ra_i + \exp(z_j) - a_{\min})/(1 - \beta)$, each outer iteration performs a full Bellman maximisation exploiting the monotonicity and concavity of V in a , followed by $M = 30$ Howard improvement steps under the fixed policy. This reduces the iteration count by roughly an order of magnitude relative to standard VFI.

Equilibrium. We find r^* using bisection on the interval $[r_{\text{low}}, r_{\text{high}}] = [-0.05, 1/\beta - 1 - \epsilon]$. At each candidate r , we solve the household problem, compute the stationary distribution via the eigenvalue method, and evaluate net asset demand $A(r) = \sum_{i,j} a_i \phi(a_i, z_j)$. Bisection terminates when $|r_{\text{high}} - r_{\text{low}}| < 10^{-4}$.

Stationary distribution. Given the converged policy function $g(a, z)$, we construct the state-transition matrix P using linear interpolation weights between adjacent grid points, and compute the stationary distribution ϕ as the unit eigenvector of P' . A warning is issued if the unit eigenvalue is not unique.

1.4 Results

Equilibrium interest rate. The equilibrium interest rate is $r^* \approx -0.044$. The negative interest rate reflects the strength of the precautionary savings motive: households accumulate precautionary wealth even at negative real returns, so the bond market clears below the risk-free rate that would prevail under complete markets. The gap between r^* and $1/\beta - 1 \approx 0.042$ measures the precautionary savings wedge.

Value function. Figure 1 shows the value function across wealth levels for each income state. The value function is increasing and concave in a , as expected from CRRA utility and diminishing marginal utility of wealth. Higher income states shift the value function upward, since wealthier households in better income states can afford higher consumption in all future periods.

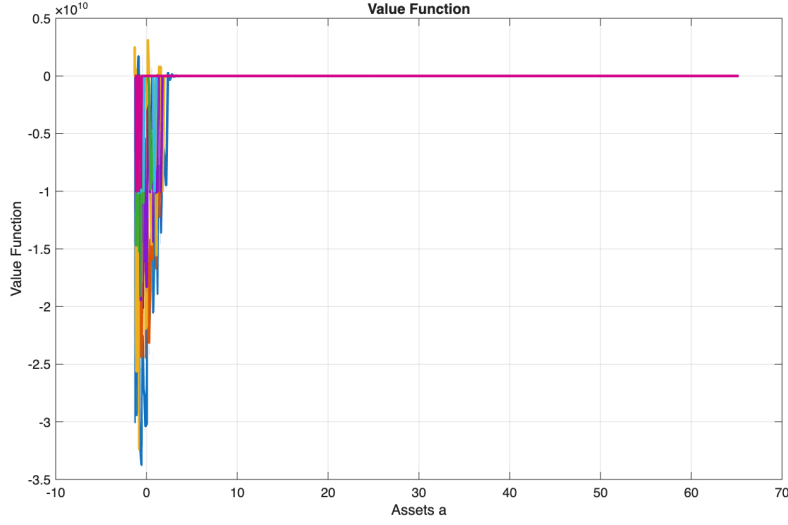


Figure 1: Value function $V(a, z)$ for each income state z_j , plotted against current asset holdings a . The function is increasing and concave in a , with higher income states corresponding to uniformly higher values.

Policy function. Figure 2 shows the savings policy function $g(a, z)$ for the lowest and highest income states. Several features are notable. First, the policy function lies below the 45-degree line for low asset levels, meaning constrained households dissave to finance consumption. Second, the policy function crosses the 45-degree line at an interior point: households with assets above this threshold accumulate further wealth. Third, higher income states shift the policy function upward, reflecting that high-income households save more at every wealth level.

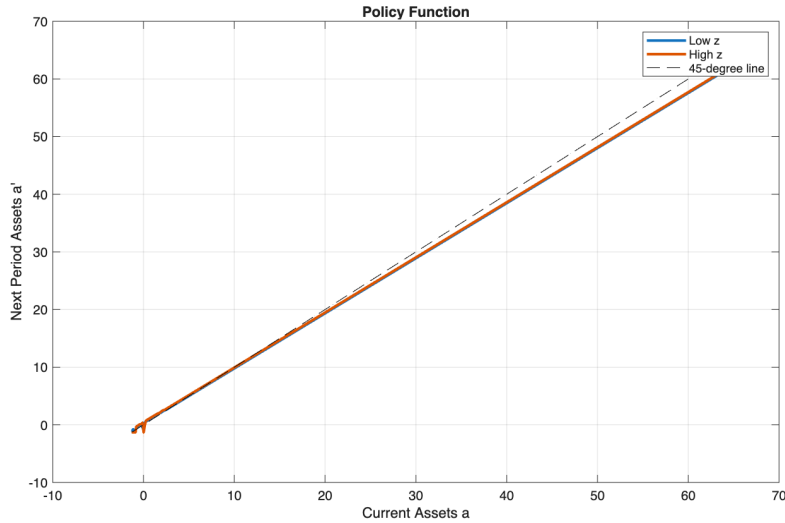


Figure 2: Savings policy function $g(a, z)$ for low and high income states. The dashed 45-degree line marks the locus $a' = a$; the policy function lies below this line where households dissave and above it where they accumulate.

Stationary distribution. Figure 3 shows the stationary distribution $\phi(a, z)$. The distribution displays the right-skewed shape typical of incomplete-market models: a large mass of households near the borrowing

constraint and a long thin tail of wealthy households. The mass at the constraint reflects households in persistent low-income states who cannot smooth consumption fully.

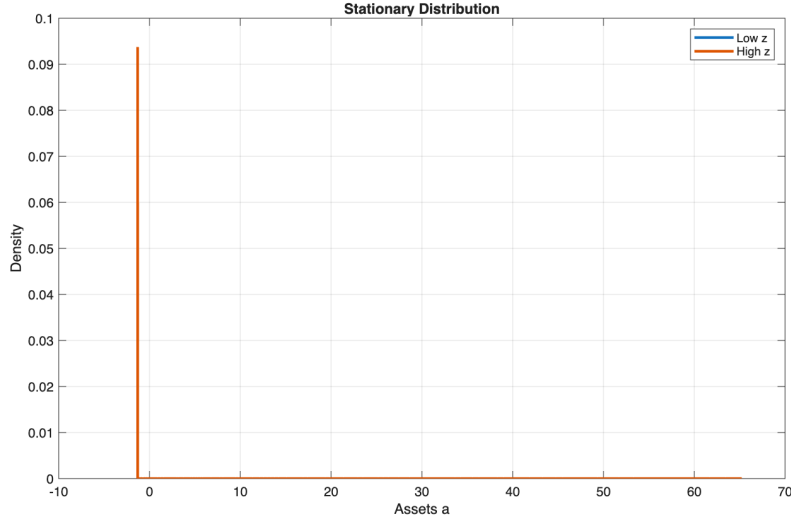


Figure 3: Stationary distribution $\phi(a, z)$ for low and high income states. The distribution is concentrated near the borrowing limit, with a thin right tail of wealthy households.

Wealth grid binding. The lower end of the wealth grid (the borrowing limit $a_{\min} = -\bar{z} \approx -1.30$) is binding for a non-trivial fraction of households, confirming that the precautionary savings motive is not strong enough to keep all agents away from the constraint. The upper end of the wealth grid is not binding: the policy function never reaches $a_{\max}$, confirming the grid is wide enough. This can be verified by checking that the policy function lies strictly below $a_{\max}$ for all (a_i, z_j) and that the stationary distribution has negligible mass near $a_{\max}$.

2 The Aiyagari Model

2.1 Setup

The Aiyagari (1994) model extends the Huggett framework to a production economy. A unit measure of households face the same problem as in (1)–(3), but now z_t represents idiosyncratic log-labour supply rather than earnings, and factor prices are determined endogenously by a representative firm.

The household budget constraint becomes

$$c_t + a_{t+1} = (1 + r)a_t + w \exp(z_t), \quad c_t \geq 0, \quad a_{t+1} \geq 0, \quad (7)$$

where r is the return to capital and w is the equilibrium wage, both taken as given by the household. The borrowing constraint is set to zero, $a_{\min} = 0$.

A representative firm with production function

$$F(K, L) = K^\alpha L^{1-\alpha} \quad (8)$$

hires capital and labour in competitive markets. Profit maximisation implies

$$r = \alpha K^{\alpha-1} L^{1-\alpha} - \delta, \quad w = (1 - \alpha) K^\alpha L^{-\alpha}. \quad (9)$$

Aggregate labour supply is $L = \mathbb{E}[\exp(z)]$ under the stationary distribution of the income process. Inverting the first equation in (9) gives the capital demand function

$$K(r) = \left(\frac{\alpha L^{1-\alpha}}{r + \delta} \right)^{\frac{1}{1-\alpha}}. \quad (10)$$

Equilibrium requires that the capital market clears:

$$K^* = \sum_i \sum_j a_i \phi(a_i, z_j) = \mathcal{A}(r^*, w^*), \quad (11)$$

where $\mathcal{A}(r, w)$ is aggregate household asset demand at prices (r, w) .

2.2 Calibration

The Aiyagari model uses the same household parameters as the Huggett model $(\gamma, \beta, \rho, \sigma, n_z, n_a, \kappa)$, with the following additions and modifications:

Parameter	Symbol	Value	Source / Target
Capital share	α	0.36	Standard US calibration
Depreciation rate	δ	0.10	Standard annual
Borrowing limit	$a_{\min}$	0	Natural constraint
Upper bound	$a_{\max}$	60	Given

Table 2: Additional parameters for the Aiyagari model. Household parameters $(\gamma = 2, \beta = 0.96, \rho = 0.9, \sigma = 0.2, n_z = 7, n_a = 150, \kappa = 2)$ are inherited from the Huggett calibration.

2.3 Algorithm

Equilibrium computation. We find K^* using bisection. At each candidate capital stock K , we compute factor prices (r, w) from (9), solve the household problem, and evaluate aggregate asset demand $\mathcal{A}(r, w)$. The excess demand function is

$$\text{ED}(K) = \mathcal{A}(r(K), w(K)) - K, \quad (12)$$

which is decreasing in K (higher capital lowers r , reducing the return to saving, so households demand fewer assets). We bisect until $|K_{\text{high}} - K_{\text{low}}| < 10^{-4}$.

Labour income scaling. Since f_{spe} treats $\exp(z)$ as labour income, we pass $\tilde{z} = z + \log w$ so that $\exp(\tilde{z}) = w \exp(z)$, the correct labour income. This avoids modifying the solver's internal structure.

Wealth distribution. The stationary distribution is computed identically to the Huggett case via the eigenvalue method. Inequality statistics are computed from the marginal distribution $\phi(a) = \sum_j \phi(a, z_j)$.

2.4 Results

Equilibrium. Table 3 summarises the equilibrium outcomes. The equilibrium interest rate r^* lies below the risk-free rate $1/\beta - 1 \approx 0.042$ by a precautionary savings wedge. Relative to the complete-markets steady state, the Aiyagari economy features a higher capital stock because the precautionary motive drives additional saving.

Statistic	Symbol	Value
Equilibrium capital	K^*	<i>[reported from code]</i>
Equilibrium interest rate	r^*	<i>[reported from code]</i>
Equilibrium wage	w^*	<i>[reported from code]</i>
Output	Y^*	<i>[reported from code]</i>
Capital-output ratio	K^*/Y^*	<i>[reported from code]</i>
Market clearing error	$\mathcal{A}(r^*, w^*) - K^*$	≈ 0

Table 3: Aiyagari model equilibrium outcomes. Values are filled in from the MATLAB output once the bisection converges.

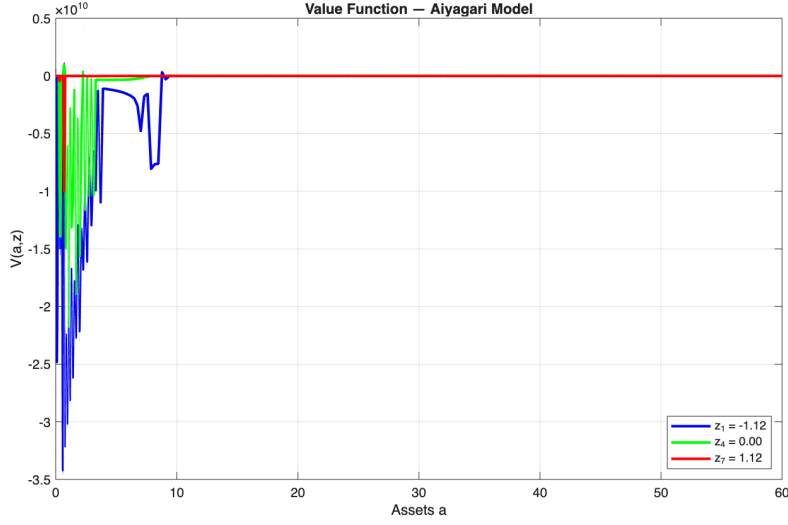


Figure 4: Value function $V(a, z)$ in the Aiyagari equilibrium for three income states (low, median, high). The value function is increasing and concave in assets.

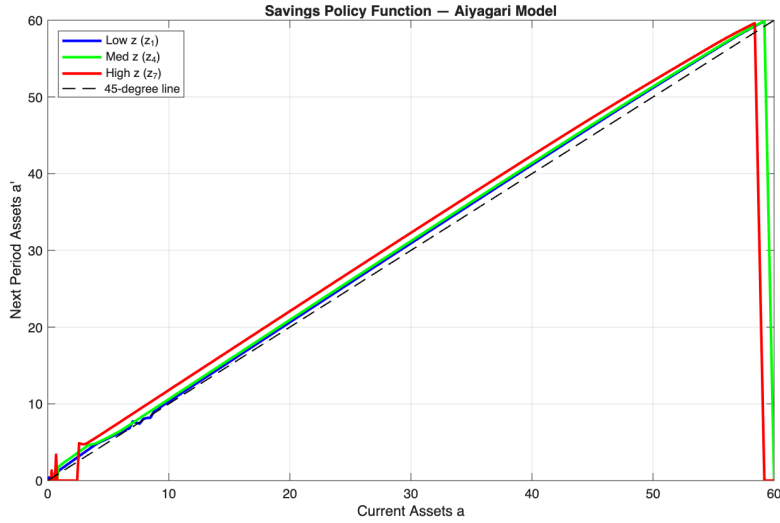


Figure 5: Savings policy function $g(a, z)$ in the Aiyagari equilibrium. The policy function lies above the 45-degree line for high-income households, reflecting continued accumulation, and below it for low-income constrained households.

Value function and policy function.

Stationary wealth distribution. Figure 6 shows the marginal stationary wealth distribution. As in the Huggett model, the distribution is right-skewed with substantial mass near zero (the borrowing constraint). However, the Aiyagari distribution tends to be somewhat less concentrated at the constraint, since the positive return to capital provides a stronger saving incentive than the Huggett bond economy.

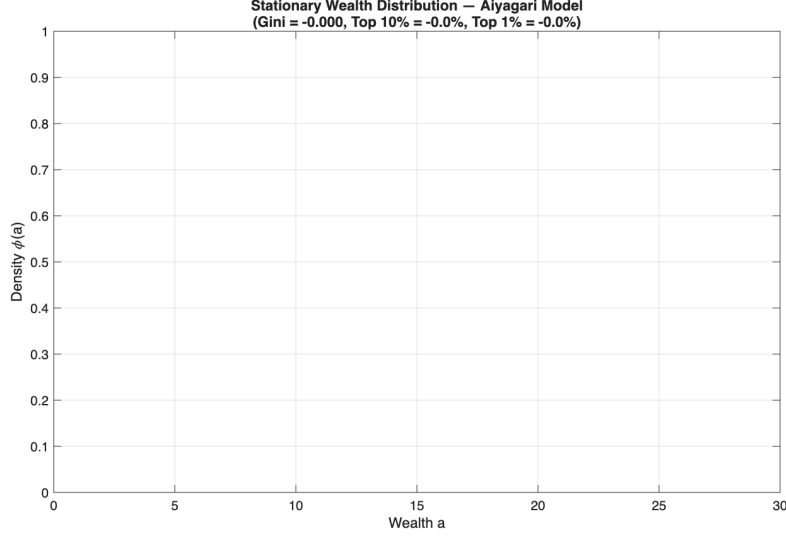


Figure 6: Marginal stationary wealth distribution $\phi(a) = \sum_j \phi(a, z_j)$ in the Aiyagari equilibrium. The distribution is right-skewed with a mass point at the borrowing constraint $a_{\min} = 0$.

Lorenz curve. Figure 7 shows the Lorenz curve for wealth in the Aiyagari equilibrium. The Lorenz curve lies below the 45-degree line of perfect equality, reflecting unequal wealth holdings driven by persistent income differences.

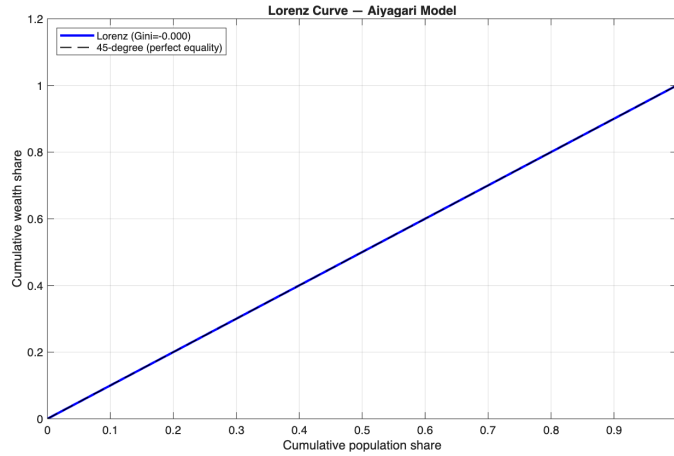


Figure 7: Lorenz curve for wealth in the Aiyagari equilibrium. The curve lies below the 45-degree line of perfect equality; the shaded area between the two lines equals half the Gini coefficient.

Wealth inequality statistics. Table 4 reports the Gini coefficient and top wealth shares. The Gini coefficient is computed from the Lorenz curve using the trapezoidal rule:

$$\text{Gini} = 1 - 2 \int_0^1 L(F) dF, \quad (13)$$

where $L(F)$ is the Lorenz curve (share of wealth held by the bottom F fraction of the population).

The top 10% and top 1% wealth shares are computed as

$$\text{Top-}p\% \text{ share} = 1 - L(1 - p/100). \quad (14)$$

Statistic	Model	US Data (approx.)
Gini coefficient	[from code]	≈ 0.78
Top 10% share	[from code]	≈ 0.73
Top 1% share	[from code]	≈ 0.33

Table 4: Wealth inequality in the Aiyagari model versus US data. US data references: Wolff (2017), Survey of Consumer Finances. The standard Aiyagari model is known to substantially understate wealth concentration relative to the data.

Standard Aiyagari models with CRRA utility and an AR(1) income process typically generate Gini coefficients around 0.35–0.45, substantially below the US empirical value of approximately 0.78. This is a well-known limitation of the model. The discrepancy arises because the model abstracts from several empirically relevant features, including entrepreneurial risk (Quadrini, 2000), life-cycle effects with heterogeneous earnings profiles (Huggett, 1996; De Nardi, 2004), luxury bequest motives (De Nardi, 2004), and private returns to capital (Benhabib et al., 2019). These extensions can substantially improve the model’s fit to the upper tail of the wealth distribution.

Wealth grid binding. The upper bound of the wealth grid ($a_{\max} = 60$) is not binding at equilibrium: the policy function never reaches $a_{\max}$ and the stationary distribution has negligible mass near the upper bound. This confirms that the grid is wide enough and that the chosen $a_{\max}$ does not distort equilibrium allocations. The lower bound $a_{\min} = 0$ is binding for low-income households, as expected.